QUALITY INFORMATION DOCUMENT

For
OMI_CLIMATE_TEMPSAL_BALTIC_Stz_trend

Issue: 4.0

Contributors : P. Lagemaa, U. Raudsepp, I. Maljutenko, V. Huess

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MERCATOR OCEAN
INTERNATIONAL
2 avenue de l'Aérodrome de Montaudran, 31400 Toulouse, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00024

marine.copernicus.eu
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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2018/09/06	all	Creation of the document	U. Raudsepp I. Maljutenko	P. Lagemaa
1.1	2019/02/21		Revision	U. Raudsepp I. Maljutenko	P. Lagemaa
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TABLE OF CONTENTS

RECORD TABLE **2**

1) INTRODUCTION **4**

a) Product Name 4

b) Indicator Evaluation Method 4

c) Copernicus Marine products used as input for the computation 4

2) SCIENTIFIC VALIDATION **5**

a) Subsurface salinity trend 5

3) REFERENCES **6**

1) INTRODUCTION

a) Product Name

OMI_CLIMATE_TEMPSAL_BALTIC_Stz_trend.

b) Indicator Evaluation Method

This document provides to Copernicus Marine Service users information on the quality of the horizontally averaged salinity trend for the Baltic Sea. The linear trends/slopes/fits and their 95% confidence intervals have been calculated using last square regression method described in Draper and Smith (1998) eq. 1.2.9 and 1.4.8.

The daily salinity data is extracted from the BALTICSEA_MULTIYEAR_PHY_003_011 product. The accuracy of the index is directly dependent on the accuracy of the Baltic reanalysis product (BALTICSEA_MULTIYEAR_PHY_003_011) therefore evaluation is based on the QUID documents of the aforementioned product.

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
BALTICSEA_MULTIYEAR_PHY_003_011 (model reanalysis)	QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-BAL-QUID-003-011.pdf PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-BAL-PUM-003-011_012.pdf

2) SCIENTIFIC VALIDATION

a) Subsurface salinity trend

In the CMEMS-BAL-QUID-003-011 the salinity of reanalysis product has been compared with all available measurements across the Baltic for period of 1993-2024. As the data assimilation is employed the model results of reanalysis show only small biases (<0.3 g/kg) compare to observations at all observed depth ranges. The RMSE showed values around 0.48 – 0.73 g/kg in the layer of 0-80 m and decreased toward the deeper layers to up to 0.28 g/kg in the layer of 80-220 m.

The linear trends/slopes/fits and their 95% confidence intervals (Figure 1) have been calculated using last square regression method described in Draper and Smith (1998) eq. 1.2.9 and 1.4.8.

Scientific evaluation can be found in the Copernicus Marine Service Ocean State Report (Guinehut et al., 2016; Mulet et al., 2018).

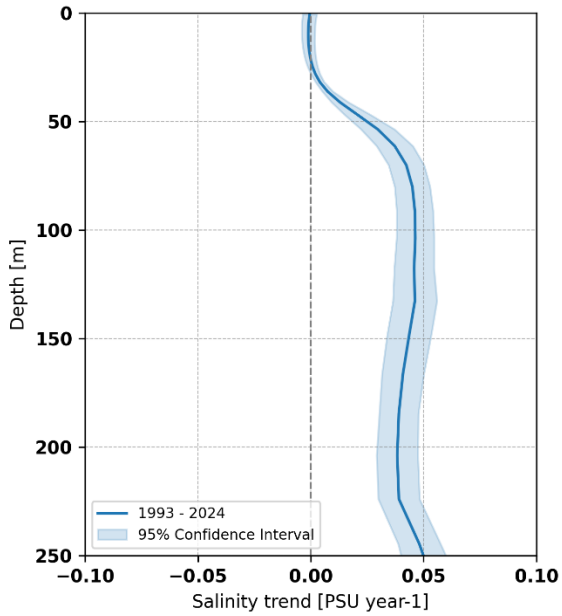

Figure 1. The vertical profile of horizontally averaged subsurface salinity trends for period of 1993-2024. The salinities from Copernicus Marine Service regional reanalysis product (BALTICSEA_MULTIYEAR_PHY_003_011) have been averaged over the Baltic Sea domain (13 °E - 31 °E; 53 °N - 66 °N; excluding the Skagerrak strait). The shaded area shows 95% confidence interval of calculated trend values.

3) REFERENCES

Draper, N. R., & Smith, H. (1998). Applied Regression Analysis. Wiley Series in Probability and Statistics. doi:10.1002/9781118625590

Guinehut et al., 2016: Surface and subsurface salinity. In: The Copernicus Marine Environment Monitoring Service Ocean State Report, issue 1, Journal of Operational Oceanography, Volume 9, 2016 - Issue sup2: The Copernicus Marine Environment Monitoring Service Ocean, http://dx.doi.org/10.1080/1755876X.2016.1273446.

Mulet, S., Nardelli, BB., Good, S., Pisano, A., Greiner, E., Monier, M., Autret, E., Axell, L., Boberg, F., Ciliberti, S., Drévillon, M., Droghei, R., Embury, O., Gourrion, J., Høyer, J., Juza, M., Kennedy, J., Lemieux-Dudon, B., Peneva, E., Reid, R., Simoncelli, S., Storto, A., Tinker, J., von Schuckmann, K., Wakelin, S.L., 2018. Ocean temperature and salinity. In: Copernicus Marine Service Ocean State Report, Issue 2, Journal of Operational Oceanography, 11:sup1, s5–s13, DOI:10.1080/1755876X.2018.1489208